

UNIVERSITY OF EDINBURGH
COLLEGE OF SCIENCE AND ENGINEERING
SCHOOL OF INFORMATICS

INFR09027 COMPUTER COMMUNICATIONS AND NETWORKS

Thursday 28th April 2016

09:30 to 11:30

INSTRUCTIONS TO CANDIDATES

Answer any TWO questions.

All questions carry equal weight.

CALCULATORS MAY NOT BE USED IN THIS EXAMINATION

Year 3 Courses

Convener: C. Stirling

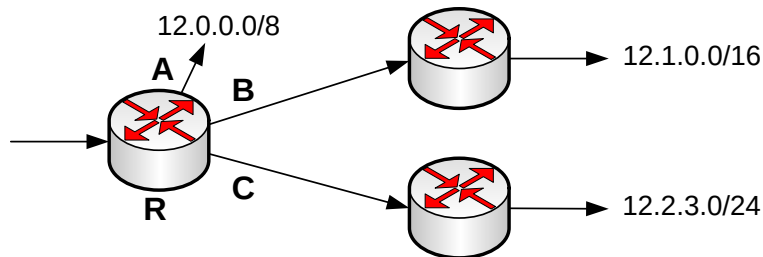
External Examiners: A. Cohn, T. Field

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

1. (a) Present two advantages and one disadvantage of packet switching over circuit switching. [3 marks]
- (b) Suppose a web site (`aaa.com`) has a base HTTP document (`index.html`) that references the following five objects: `foo.com/a.jpg`, `foo.com/b.jpg`, `foo.com/c.jpg`, `bar.com/d.jpg` and `bar.com/e.jpg`. Assume that RTT (round-trip time) is the same from the client to servers `foo.com` and `bar.com`, and capacity between the client and two servers is infinite. The browser supports parallel TCP connections.
 - i. Assume the browser uses a non-persistent HTTP connection. How many RTTs does it take to download all of the objects (including the base HTTP document)? Explain your answer. [2 marks]
 - ii. Assume the browser uses a persistent HTTP connection, but serialises HTTP request and reply messages. This means that within a single TCP connection a new request can be sent once the reply message for the previous request arrives at the browser. Under these conditions, answer the same question in part i. [2 marks]
- (c) An HTTP response message can include a “Last Modified” time, indicating the last time the requested object was modified. When a user requests an object that resides in the browser cache, the browser generates an HTTP request message with an “If Modified Since” header to ask the server to return a fresh copy only if the object has been modified since the previous request. How is it possible for the “If Modified Since” cache-validation technique to work, even if the two machines have vastly different notions of time (i.e., no clock synchronisation)? [5 marks]
- (d) List three key reasons for a host to have both a domain name and an IP address. [3 marks]
- (e) BitTorrent, a P2P file sharing protocol, employs a “tit-for-tat” strategy.
 - i. Explain the strategy briefly. [2 marks]
 - ii. In the strategy, a user “optimistically unchokes” its peers. Why is the optimistically-unchoking mechanism necessary? Present two reasons. [4 marks]
- (f) An office has an (Internet) access link rated at 10 Mbps full-duplex. The office supports 4 users on the packet-switched access link. Each user requires 5 Mbps when transmitting and each user is active 20% of the time. Assuming the users are independent, find the probability that there are 3 or more users transmitting simultaneously. Given that probability, is packet-switching a better option than circuit-switching? Justify your answer! [4 marks]

2. (a) In TCP demultiplexing, a 4-tuple <source IP, source port number, destination IP, destination port number> is necessary. In other words, a socket is identified by the 4-tuple. What can go wrong if the source port number, destination IP and destination port number are used only in the TCP demultiplexing? Give a specific example. [3 marks]
- (b) Host A is transferring a file to host B over a path with a RTT of 100ms and a bottleneck bandwidth of 100Kbps. Host A has a fixed sliding window of 1KB. What is the maximum throughput Host A can achieve to Host B, ignoring packet headers? [2 marks]
- (c) Briefly describe three costs of congestion. [3 marks]
- (d) Briefly explain the additive increase multiplicative decrease (AIMD) algorithm. [2 marks]
- (e) A sender and a receiver are communicating using a TCP-based protocol in which data is sent immediately after connection establishment (i.e., the first data payload is sent with the final ACK in the three-way handshake of connection establishment). The bandwidth of the link is 100Mbps, the round-trip time is 10ms and the maximum packet/segment size is 1,000 bytes (ignore header overhead). Assume the initial congestion window is set to one MSS (maximum segment size), that the receiver has ample buffer space and the initial value of `ssthresh` is high (e.g., 64,000 bytes). Assume default values for TCP parameters.
 - i. How many RTTs does it take to transfer 7,000 bytes? Show your work. [5 marks]
 - ii. The fast retransmit optimisation is used to avoid a timeout. Consider the TCP transfer in the previous part i. above. If packets were lost during that transfer, could fast retransmit ever come into play? If so, which packet loss could trigger the fast retransmit? [5 marks]
- (f) Suppose after sending a series of packets, $1..n$, TCP does not receive any acknowledgements for an extended period of time. Normally, after a certain timeout period is reached, TCP will reset the congestion window to 1, and will retransmit the first packet (packet 1). Suppose instead, that TCP were modified to send the $(n + 1)^{th}$ packet in this case. Under what conditions would this be a good idea and why? (Be specific.) Under what circumstances would this be a bad idea and why? (Be specific.) [5 marks]

3. (a) Briefly describe three different scenarios where an IP router would drop packets that arrive on one of its interfaces. [3 marks]
- (b) Consider the following three routers, where router R has outgoing interfaces A, B, and C:



- i. How many addresses does the IP prefix 12.2.3.0/24 represent? [2 marks]
 - ii. List the forwarding-table entries for router R. How does R perform look-ups in this table? [3 marks]
 - iii. Which outgoing interface does R use for a packet with destination 12.2.3.1? [2 marks]
 - iv. Which outgoing interface does R use for a packet with destination 12.2.4.5? [2 marks]
- (c) List three key differences between MAC and IP addresses. [3 marks]
- (d) Suppose the Internet is constructed using switches (link-layer devices) only. This means that packet forwarding is purely based on MAC address instead of IP address. That is, the forwarding table consists of a list of 2-tuple <MAC address, outgoing interface number> entries. What would be a key problem of such a design? How can the use of IP addresses overcome this problem? [4 marks]
- (e) Why do Ethernet adaptors select a *random* back-off time before trying to transmit a frame following a collision? Why do they pick the random back-off time from a *larger range* after each collision? [6 marks]